SADIG AKHUND

Data Engineer

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Data Engineer with 3+ years across platform and delivery — integrated **Apache Spark, Iceberg, and Trino** into an in-house orchestration platform, and shipped lakehouse pipelines processing tens of millions of records for government and financial clients.

Experience

KDM Force
Data Engineer (Remote)

Rome, Italy
Jun 2022 – Nov 2025

Platform Engineering — Fyrefuse orchestration platform

- Researched and integrated **Apache Iceberg, Hive Metastore, Trino, PostgreSQL, and Apache Spark** into the in-house Fyrefuse platform, opening federated lakehouse queries over object storage without re-ingesting to the warehouse.
- Built a **viewport-aware caching layer** between the application frontend and Trino-backed sources — preloaded pages around the user's scroll window in both directions, eliminating perceived load time on multi-million-row CSV-style tables.
- Selected **PostgreSQL** over Redis as the cache backend after benchmarking: Redis covered only key-value access while users needed sort, filter, aggregation, and full-text search; Postgres delivered all four plus **sub-second write times for 1M records** after switching the ingest path from `INSERT`/bulk-insert to `COPY`.

Delivery — Ministry of Tourism of Saudi Arabia

- Designed multi-source OTA ingestion: extraction scripts published to **Apache Kafka** on standardized topics (decoupling ingest from transform, enabling replay), consumed by **Spark** pipelines orchestrated through Fyrefuse, landing in a **MS SQL Server** warehouse.
- Modeled the warehouse schema for **80M+ customer reviews** and multi-year listing metadata across **18 business-critical tables**, applying dimensional modeling and slowly-changing-dimension patterns to power national tourism analytics.
- Engineered adaptive error handling for unreliable OTA providers (rate-limit backoff, schema drift detection, dead-letter queue), eliminating routine manual intervention.
- Owned ETL workflow governance in **Dataiku** — parameter file management, log auditing, pipeline observability across all ingestion jobs.

Stack: Apache Spark, Kafka, Iceberg, Trino, Hive Metastore, PostgreSQL, Redis, MS SQL Server, Dataiku, RHEL

Unibank
Data Engineer (Intern)

Baku, Azerbaijan
Oct 2021 – May 2022

- Studied and documented the bank's production **push-notification telemetry pipeline** (mobile app → **Apache NiFi** ingestion → **Apache Spark** enrichment → **ELK stack**), surfacing schema gaps and partition hotspots to the platform team.
- Refactored middleware log schema feeding Logstash, cutting per-event payload size and standardizing the format consumed by Spark enrichment jobs — reduced storage cost and improved Kibana query latency.
- Built a **FastAPI** service that exposed enriched notification metrics to internal stakeholders, replacing manual ad-hoc Elasticsearch queries.
- Benchmarked **InfluxDB, Snowflake, and MongoDB** as candidate stores for time-series and document workloads, producing a feasibility report that informed the bank's analytics roadmap.

Stack: Apache Spark, Apache NiFi, Elasticsearch, Logstash, Kibana, Python, FastAPI, MongoDB, InfluxDB, Snowflake

Education

George Washington University 2021 – 2023
M.Sc., Computer Science & Data Analytics

ADA University 2017 – 2021
B.Sc., Computer Engineering

Projects

Aqueduct · Declarative Spark engine with agentic self-healing GitHub — Feb 2026 – Present

- Designed and shipped a **declarative pipeline engine** (Parser → Compiler → Executor → Surveyor) replacing imperative PySpark with version-controlled YAML Blueprints. Released **1.0.0 to PyPI** as `aqueduct-core`.
- Built a multi-provider **LLM self-healing agent loop** that diagnoses failed Spark runs from a structured `FailureContext` and emits validated patches via a **10-op PatchSpec grammar**; deterministic guardrails + provenance-aware prompts cut hallucination.
- Shipped **first-class Apache Airflow integration**: drop-in `AqueductOperator` + deferred-pattern sensors; supports human/CI healing modes via structured exit codes. Replaces the imperative-Spark layer underneath existing orchestration.
- Engineered **zero-overhead observability** — pipeline health, performance metrics, and column-level data lineage tracked automatically with no measurable performance cost, built on Spark's native instrumentation hooks and an embedded analytical database.
- Implemented **pluggable persistence** (DuckDB, Postgres, Redis behind a Store API), multi-cloud secrets dispatch (AWS / GCP / Azure), structured Spark error extraction, and a 2-tier benchmark harness for prompt regression detection in CI.

Acquired Skills: Apache Spark, Apache Airflow, DSL & Compiler Design, LLM Agents, Prompt Engineering, DuckDB, Data Lineage, Platform Engineering, Open-Source Maintainership

Certifications

Google Data Analytics Professional Certificate · Verify on Credly — Apr 2022

Skills

Data Engineering	Apache Spark, Kafka, Airflow, dbt, Iceberg, Delta Lake, Trino, ETL/ELT
Languages	Python, SQL, Java, C++, Bash
Databases	PostgreSQL, Snowflake, BigQuery, DuckDB, MS SQL Server, MongoDB, Redis
Cloud & DevOps	GCP, AWS, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Linux
LLM & Agents	Anthropic, OpenAI, Ollama, agent loops, prompt engineering, structured output
Backend & Tools	FastAPI, Django, Pydantic, Click, sqlglot, Git, REST APIs
Analytics & BI	Looker, Superset, Grafana, Metabase, Data Modeling
Spoken Languages	English (Proficient), Russian (Advanced), Azerbaijani (Native)